

**NORME
INTERNATIONALE**

**CEI
IEC**

**INTERNATIONAL
STANDARD**

60853-3

Première édition
First edition
2002-02

**Calcul des capacités de transport des câbles
pour les régimes de charge cycliques et
de surcharge de secours –**

**Partie 3:
Facteur de capacité de transport cyclique
pour des câbles de toute tension, avec
assèchement partiel du sol**

**Calculation of the cyclic and emergency
current rating of cables –**

**Part 3:
Cyclic rating factor for cables of all voltages,
with partial drying of the soil**



Numéro de référence
Reference number
CEI/IEC 60853-3:2002

**NORME
INTERNATIONALE
INTERNATIONAL
STANDARD**

**CEI
IEC**

60853-3

Première édition
First edition
2002-02

**Calcul des capacités de transport des câbles
pour les régimes de charge cycliques et
de surcharge de secours –**

**Partie 3:
Facteur de capacité de transport cyclique
pour des câbles de toute tension, avec
assèchement partiel du sol**

**Calculation of the cyclic and emergency
current rating of cables –**

**Part 3:
Cyclic rating factor for cables of all voltages,
with partial drying of the soil**

© IEC 2002 Droits de reproduction réservés — Copyright - all rights reserved

Aucune partie de cette publication ne peut être reproduite ni utilisée sous quelque forme que ce soit et par aucun procédé, électronique ou mécanique, y compris la photocopie et les microfilms, sans l'accord écrit de l'éditeur.

No part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from the publisher.

International Electrotechnical Commission, 3, rue de Varembe, PO Box 131, CH-1211 Geneva 20, Switzerland
Telephone: +41 22 919 02 11 Telefax: +41 22 919 03 00 E-mail: inmail@iec.ch Web: www.iec.ch



Commission Electrotechnique Internationale
International Electrotechnical Commission
Международная Электротехническая Комиссия

**CODE PRIX
PRICE CODE**

N

*Pour prix, voir catalogue en vigueur
For price, see current catalogue*

CONTENTS

FOREWORD.....	5
INTRODUCTION.....	7
1 Scope.....	9
2 Normative references	9
3 Symbols	11
4 Description of method.....	13
4.1 General description	13
4.2 Principles	13
4.3 Formulae.....	15
Annex A (informative) Example calculation	21
Annex B (informative) Details of cables and load used for example.....	27

INTERNATIONAL ELECTROTECHNICAL COMMISSION

**CALCULATION OF THE CYCLIC
AND EMERGENCY CURRENT RATING OF CABLES -****Part 3: Cyclic rating factor for cables of all voltages,
with partial drying of the soil**

FOREWORD

- 1) The IEC (International Electrotechnical Commission) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of the IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, the IEC publishes International Standards. Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. The IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of the IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested National Committees.
- 3) The documents produced have the form of recommendations for international use and are published in the form of standards, technical specifications, technical reports or guides and they are accepted by the National Committees in that sense.
- 4) In order to promote international unification, IEC National Committees undertake to apply IEC International Standards transparently to the maximum extent possible in their national and regional standards. Any divergence between the IEC Standard and the corresponding national or regional standard shall be clearly indicated in the latter.
- 5) The IEC provides no marking procedure to indicate its approval and cannot be rendered responsible for any equipment declared to be in conformity with one of its standards.
- 6) Attention is drawn to the possibility that some of the elements of this International Standard may be the subject of patent rights. The IEC shall not be held responsible for identifying any or all such patent rights.

International Standard IEC 60853-3 has been prepared by IEC technical committee 20: Electric cables.

The text of this standard is based on the following documents:

FDIS	Report on voting
20/493/FDIS	20/512/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 3.

Annexes A and B are for information only.

The committee has decided that the contents of this publication will remain unchanged until 2007. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

INTRODUCTION

Methods for calculating cyclic ratings for cables installed in the ground are given in IEC 60853-1 and IEC 60853-2. Part 1 is applicable to cables having negligible thermal capacitance; Part 2 is applicable to cables where their thermal capacitance cannot be neglected.

A fundamental assumption in Part 1 and Part 2 is that the thermal properties of the soil are uniform throughout the thermal field of the cable. For many installations, such an assumption is acceptable and further refinement is unnecessary.

However, higher working temperatures increase the probability that the soil immediately around a cable may lose so much of its moisture that it dries out to a significantly higher value of thermal resistivity than that corresponding to its normal site condition. The reduction in permissible current-carrying capacity of a cable, necessary to avoid overheating, can be significant.

CALCULATION OF THE CYCLIC AND EMERGENCY CURRENT RATING OF CABLES -

Part 3: Cyclic rating factor for cables of all voltages, with partial drying of the soil

1 Scope

This part of IEC 60853 gives a method for calculating the cyclic rating factor, for cables of all voltages, where partial drying out of the surrounding soil is anticipated. The method is based on one of the three methods published in a CIGRE¹ document. The method is applicable to any type of cable, but it is recommended that it should be applied only to installations of one multi-core cable or to three single-core cables or to groups of circuits where the intercircuit spacing is sufficient to permit free vertical movement of soil moisture between the zones of dry soil associated with each circuit. This standard does not preclude the use of other methods of calculation where full details of the load cycle are not known.

The method assumes that the entire region surrounding a cable or cables has uniform thermal characteristics prior to drying out; the only non-uniformity being that caused by drying. As a consequence the method should not be applied, without further consideration, to installations where special backfills, having properties different from the site soil, are used.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60287-2-1, *Electric cables - Calculation of the current rating - Part 2-1: Thermal resistance - Calculation of thermal resistance*

IEC 60853-1, *Calculation of the cyclic and emergency current rating of cables - Part 1: Cyclic rating factor for cables up to and including 18/30 (36) kV*

IEC 60853-2:1989, *Calculation of the cyclic and emergency current rating of cables - Part 2: Cyclic rating of cables greater than 18/30 (36) kV and emergency ratings for cables of all voltages*

¹ Methods for calculating cyclic ratings for buried cables with partial drying out of the surrounding soil: *Electra* No 145, December 1992, page 32.

3 Symbols

a, b	coefficients used for calculating cable partial temperature rise, s^{-1}
e	2,7183
i	time, in hours (h)
k	steady-state ratio of cable surface temperature rise to conductor temperature rise, above ambient
n	number of cores
t	time, in seconds (s)
v	ratio of dry to wet soil thermal resistivities ($v = \rho_d / \rho_w$)
A, A', B	intermediate variables, used in 4.3.1 and 4.3.2
D_e	cable external diameter, in meters (m)
Ei	exponential integral function
L	depth a cable is laid below ground surface, in meters (m)
M	cyclic rating factor
M_1	cyclic rating factor corrected for moisture migration
Q_A, Q_B	elements of two part thermal circuit of cable, J/m·K
T_a, T_b	apparent thermal resistances used to calculate cable partial transient, K·m/W
T_A, T_B	elements of equivalent thermal circuit of cable, K·m/W
T_1	thermal resistance of cable insulation per conductor, K·m/W
T_3	thermal resistance of cable covering, K·m/W
T_4	external thermal resistance of cable, K·m/W
W_d	dielectric losses per cable, W/m
Y_i	cable losses, expressed as a fraction of their maximum value, at a time i (h) prior to the time of maximum conductor temperature, W/m
$\alpha(t)$	attainment factor of temperature difference between cable surface and conductor
$\beta(t)$	attainment factor of temperature difference between ambient and the cable surface
δ	soil thermal diffusivity, m^2/s
λ	armour/sheath loss factor
μ	loss load factor
ρ_w	soil thermal resistivity for moist soil, K·m/W
ρ_d	soil thermal resistivity for dry soil, K·m/W
θ_c	conductor temperature rise due to total cable losses, in kelvins (K)
θ'_c	conductor temperature rise due to joule losses only, in kelvins (K)
$\theta'_c(i)$	conductor temperature rise at time i (h), in kelvins (K)
θ_x	critical value of soil temperature rise above ambient, in kelvins (K)

NOTE All temperature symbols refer to rises above ambient.

4 Description of method

4.1 General description

This method is an extension of the techniques used in IEC 60853-1 and IEC 60853-2 for calculating cyclic rating factors for a cable in uniform soil.

The method relies on a knowledge of the soil critical temperature; this is the temperature at which drying out of the soil takes place rapidly.

NOTE In the absence of more precise information, this temperature may be taken as 50 °C.

The size of the dry zone changes substantially with critical temperature. For a given conductor temperature the size of the dry zone will have a significant effect on the steady-state rating but the cyclic factor multiplying that steady-state rating is largely unaffected.

It is assumed initially that the soil critical temperature is equal to the peak cyclic value of the cable surface temperature, i.e. a dry zone is on the point of occurring, so that the soil surrounding the cable has uniform properties appropriate to its wet, on-site, state. The cyclic rating factor for these conditions is derived by the methods already given in IEC 60853-1 or IEC 60853-2.

This factor is then adjusted so that it applies to the steady-state rating for the same assumed value of critical temperature, when there will be a dried-out zone. This adjustment is effected using formulae for the cable external thermal resistance appropriate to the steady-state procedure.

Such a factor can then be used to multiply the steady-state rating for any other soil critical temperature in order to obtain the appropriate permissible peak current.

4.2 Principles

In general, the size of a dry zone where the boundary just achieves a certain critical temperature rise with cyclic loading is smaller than the zone which will form for the same critical temperature rise with steady-state loading. The situation where the cable surface temperature rise is just equal to the soil critical temperature, and a dry zone forms only with steady-state loading, is a particular case. However, a rating factor determined for this latter case is applicable to the steady-state rating with any other critical temperature rise and size of dry zone.

One further step is necessary in order to use a cyclic factor based on a critical temperature equal to the cable surface temperature. Because of the nature of the computation used in IEC 60853-1 and IEC 60853-2 to derive cyclic rating factors, the value of the factor obtained assumes that the cable external thermal resistance is the same for both cyclic and steady-state loading. The correct value of peak current for the load cycle is obtained when the cyclic factor multiplies the steady-state rating for this value of resistance.

While this equality of external thermal resistance applies for the uniform non-migration conditions assumed in IEC 60287-2-1, IEC 60853-1 and IEC 60853-2, it is not so when drying can take place. The size of the dry zone, and hence the cable external thermal resistance, changes with the type of loading. In the latter case the rating factor has to be adjusted so that it can be used to multiply the rating for the higher external thermal resistance occurring with the steady-state. Such an adjustment can be made by using the ratio of the appropriate external thermal resistances.

rature

terne,
53-2;
éant)

The computation of a cyclic rating factor, for any critical temperature, is effected as follows:

- the thermal response of the cable, including the effect of its internal thermal capacitance, is obtained by the use of appropriate formulae from 4.2.1, 4.2.2 and 4.2.3 of IEC 60853-2;
- the thermal response of the soil, and the effect of adjacent cables in a group (if any), are obtained by the use of formulae from clauses 5 to 7 of IEC 60853-2;
- the effect of drying out is obtained by the use of 4.3.2.

4.3 Formulae

4.3.1 Without drying out

acité

There are optional formulae for a cyclic rating factor in clause 5 of IEC 60853-2 but the general one is as follows:

(1)

$$M = \frac{1}{(A+B)^{0.5}} \quad (1)$$

(2)

where

$$A = \sum_{i=0}^5 Y_i \left[\frac{\theta'_c(i+1)}{\theta'_c(\infty)} - \frac{\theta'_c(i)}{\theta'_c(\infty)} \right] \text{ with } \frac{\theta'_c(0)}{\theta'_c(\infty)} = 0 \quad (2)$$

(3)

and

$$B = \mu \left[1 - \frac{\theta'_c(6)}{\theta'_c(\infty)} \right] \quad (3)$$

4)

$$\frac{\theta'_c(t)}{\theta'_c(\infty)} = [1 - k + k\beta(t)]\alpha(t) \quad (4)$$

es
es
es

NOTE 1 The temperature rises referred to here are mostly those due to the joule losses only, and the equations quoted take this into account at the appropriate stages. A symbol followed by a value or a symbol in parentheses indicates a time varying quantity and a prime (') over a symbol indicates a quantity which is related to joule losses only.

)

$$\beta(t) = \frac{-Ei \left(\frac{-D_e^2}{16\delta t} \right)}{2 \cdot \ln \left(\frac{4L}{D_e} \right)} \quad (5)$$

NOTE 2 The term representing the effect of the cable image is omitted from equation (5) because periods up to 6 h only are being considered.

$$\alpha(t) = \frac{T_a(1 - e^{-at}) + T_b(1 - e^{-bt})}{(T_A + T_B)} \quad (6)$$

$$k = \frac{(1 + \lambda)T_4}{T_A + T_B + (1 + \lambda)T_4} \quad (7)$$

where

λ is the sheath/armour loss factor;

μ is the loss load factor for the particular load cycle.

For cables where the internal thermal capacitance is not important, notionally cables for voltages up to and including 18/30(36) kV, the equations for M are considerably simpler and can be found in IEC 60853-1.

4.3.2 With drying out

The computation to deal with the case of possible drying out is as follows:

- a) the factor M is adjusted for the presence of a dry zone under steady-state loading conditions. This is done by calculating the ratio of the cable surface temperature rise to the conductor temperature rise with a cyclic load using equation (8).

$$(8) \quad \frac{\theta'_{SPK}}{\theta'_c} = 1 - M^2(1 - k)(A' + \mu(1 - \alpha(6))) \quad (8)$$

where

θ'_c is the permissible steady-state temperature rise of the conductor due to joule losses, in kelvins (K);

θ'_{SPK} is the peak cyclic temperature rise of the cable surface, in kelvins (K);

M is the cyclic rating factor derived from equation (1).

$$(9) \quad A' = \sum_{i=0}^5 Y_i [\alpha(i+1) - \alpha(i)] \quad (9)$$

$$(10) \quad \theta'_c = \theta_c - W_d \left(\frac{T_A}{2} + T_3 + T_4 \right) \quad (10)$$

where

θ_c is the permissible maximum conductor temperature rise due to total cable losses, in kelvins (K);

W_d is the dielectric loss for the cable under consideration, in W/m;

T_A is the thermal resistance of the cable, or equivalent cable, dielectric, in K·m/W;

T_3 is the thermal resistance of the cable covering, in K·m/W;

T_4 is the external thermal resistance of the cable assuming wet soil, in K·m/W.

NOTE The procedure for representing a three-core cable as an equivalent single core cable is set out in IEC 60853-2.

- b) the cable surface peak temperature rise shall be checked against the critical temperature rise of the soil θ_x before going further, in order to assess the possibility of drying out:
- if $(\theta'_{SPK} + W_d T_4)$ is not greater than the critical temperature rise of the soil, then there will be no drying and the cyclic rating factor is given by M without any correction;
 - if the peak temperature rise of the cable surface $(\theta'_{SPK} + W_d T_4)$ is greater than the critical value, then the corrected value, M_1 , has to be calculated.
- c) the corrected value is calculated as follows:

(11)

$$M_1 = M \left[\frac{1 + k(v - 1)}{1 + \left(\frac{\theta'_{SPK}}{\theta'_c} \right) (v - 1)} \right]^{0,5} \quad (11)$$

where

v is equal to ρ_d / ρ_w ;

ρ_w is the thermal resistivity of moist soil, in K·m/W;

ρ_d is the thermal resistivity of dry soil, in K·m/W.

For cables where the internal thermal capacitance is not important, notionally cables for voltages up to and including 18/30(36) kV, equation (11) to modify the value of M remains unchanged, but the right-hand side of equation (8) simplifies to $1 - M^2(1 - k)Y_0$. For these cables, the dielectric losses can usually be ignored and hence equation (10) becomes $\theta'_c = \theta_c$.

Annex A
(informative)

Example calculation

The example chosen is that of a three-core 400 mm² copper, paper-insulated, 132 kV oil-filled, lead sheathed and PVC covered cable.

This cable is used to illustrate the method for dealing with dried-out soil; it is described in detail in annex B.

The principal details required for the calculation of the cyclic rating factor are as follows:

$\theta_c = 65 \text{ K (above an ambient of } 20 \text{ }^\circ\text{C)}$	$D_e = 0,109 \text{ m}$
$T_1 = 0,835 \text{ K}\cdot\text{m/W}$	$T_3 = 0,09 \text{ K}\cdot\text{m/W}$
$\lambda = 0,135$	$R_c = 0,000\ 0615 \ \Omega/\text{m}$

The equivalent single-core cable – see 4.2.1.2 and 4.2.2.2 of IEC 60853-2 – has the following parameters:

$T_A = 0,278 \text{ K}\cdot\text{m/W}$	$T_B = 0,1021 \text{ K}\cdot\text{m/W}$	$W_d = 6,03 \text{ W/m per cable}$
$Q_A = 11\ 352 \text{ J/K}\cdot\text{m}$	$Q_B = 6\ 700 \text{ J/K}\cdot\text{m}$	

The calculation of cable partial transients according to 4.2.3 of IEC 60853-2 results in the following figures:

$a = 0,00\ 209$	$b = 0,000\ 221$
$T_a = 0,002\ 15$	$T_b = 0,3782$

Soil data:

$\rho_w = 1 \text{ K}\cdot\text{m/W}$	$\rho_d = 2,5 \text{ K}\cdot\text{m/W}$	$\theta_x = 30 \text{ K}$
$\delta = 0,000\ 0005 \text{ m}^2/\text{s}$	Depth of laying, $L = 1 \text{ m}$	

Load characteristics are given in annex B; the loss load factor (μ) is 0,504.

The cable response $\alpha(t)$ is given by equation (6) and is as follows:

$$\alpha(t) = \frac{0,00215(1 - e^{-0,00209t}) + 0,3782(1 - e^{-0,000221t})}{0,278 + 0,1022}$$

The thermal response of the soil is obtained from equations (5) and (7):

where

$$T_4 = \frac{1 \times \ln\left(\frac{4 \times 1}{0,109}\right)}{2\pi} = 0,5734 \text{ K}\cdot\text{m}/\text{W}$$

$$k = \frac{(1 + 0,135) \times 0,5734}{0,278 + 0,1022 + (1 + 0,135) \times 0,5734} = 0,631$$

$$\beta(t) = \frac{-Ei\left(\frac{(-0,109^2)}{16 \times t \times 0,5 \times 10^{-6}}\right)}{2 \ln\left(\frac{4 \times 1}{0,109}\right)}$$

The cyclic rating factor is given by equations (1), (2) and (3).

The following values are obtained for the time dependent items:

Time period	$\alpha(t)$	$\beta(t)$	$\frac{\theta'_c(t)}{\theta'_c(\infty)}$
1	0,5513	0,0946	0,2362
2	0,7975	0,1662	0,3778
3	0,9086	0,2137	0,4577
4	0,9588	0,2491	0,5044
5	0,9814	0,2774	0,5337
6	0,9916	0,3008	0,5540

The value of M is 1,218, from 5.2.1 of IEC 60853-2.

Up to this point, the calculations are those already given in IEC 60853-2 for the cyclic rating factor with uniform soil.

What follows is the additional computation required to deal with possible drying out.

The ratio of peak cable surface temperature rise to conductor temperature rise, from equation (8) is

$$\frac{\theta'_{SPK}}{\theta'_c} = 1 - 1,218^2 (1 - 0,631) (A' + 0,504 (1 - 0,9916)) = 0,534$$

with $A' = 0,849$ from equation (9).

The conductor temperature rise due to joule losses, from equation (10), is

$$\theta'_c = 65 - 6,03 \left(\frac{0,278}{2} + 0,090 + 0,5734 \right) = 60,2 \text{ K}$$

and the peak cable surface temperature rise due to both joule losses and dielectric losses is

$$60,2 \times 0,534 + 6,03 \times 0,5734 = 35,6 \text{ K}$$

This is greater than the critical value of 30 K, so that drying can be expected. The corrected rating factor when drying is expected is, from equation (11),

$$M_1 = 1,220 \left[\frac{1 + 0,631(2,5 - 1)}{1 + 0,534(2,5 - 1)} \right]^{0,5} = 1,27$$

The steady-state rating with drying out when the critical temperature rise is 30 K is 520 A, so that the peak current is $1,27 \times 520 = 640 \text{ A}$.

Annex B (informative)

Details of cables and load used for example

B.1 Cable and installation data

This is a three-core, 400 mm² copper conductor, 132 kV oil-filled, lead sheathed, reinforced and PVC covered cable.

Cable dimensions

Conductor diameter	22,85 mm
Conductor screen thickness	0,14 mm
Insulation thickness	9,5 mm
Insulation screen thickness	0,14 mm
Core diameter	42,4 mm
Diameter over laid up cores	91,60 mm
Diameter over binder tape	92,40 mm
Diameter over lead sheath	97,60 mm
Diameter over reinforcement	99,20 mm
Diameter over PVC oversheath	109,00 mm

Volumetric dimensions ²

Conductors	$1230 \times 10^{-6} \text{ m}^3/\text{m}$
Insulation and screens	$3008 \times 10^{-6} \text{ m}^3/\text{m}$
Steel oil ducts	$56 \times 10^{-6} \text{ m}^3/\text{m}$
Oil in ducts	$762 \times 10^{-6} \text{ m}^3/\text{m}$
Fillers	$1661 \times 10^{-6} \text{ m}^3/\text{m}$
Lead sheath	$776 \times 10^{-6} \text{ m}^3/\text{m}$
Reinforcement	$247 \times 10^{-6} \text{ m}^3/\text{m}$
PVC oversheath	$1602 \times 10^{-6} \text{ m}^3/\text{m}$

Electrical quantities

Conductor a.c. resistance at working temperature	$61,5 \times 10^{-6} \Omega/\text{m}$
Sheath loss factor	0,135
Dielectric loss	2,01 W/m per core

² The volumetric dimensions refer to the volume of each material in a one metre length of cable. These dimensions are used to calculate the thermal capacitance of each cable component.

Thermal quantities

Conductor maximum temperature	85 °C
Ambient temperature	20 °C
Conductor temperature rise	65 K
Thermal resistance of insulation, per core	0,835 K·m/W
Thermal resistance of covering	0,09 K·m/W

Thermal capacitances

Conductors	4244 J/K·m
Insulation and screens	6015 J/K·m
Fillers and oil ducts	4813 J/K·m
Sheath	1125 J/K·m
Reinforcement	940 J/K·m
PVC oversheath	2724 J/K·m
Total thermal capacitance	19861 J/K·m

Laying and soil data

Depth of laying	1,0 m
Soil moist resistivity	1,0 K·m/W
Soil dry resistivity	2,5 K·m/W
Soil moist diffusivity	0,000 0005 m ² /s
Soil dry diffusivity	0,000 000 2717 m ² /s

B.2 Load cycle

Where details of the load cycle are used in the example the last six hourly ordinates of the loss-load cycle are the only ones used. These are

Period before time of maximum conductor temperature h	Ordinate of loss-load cycle
0 – 1	0,992
1 – 2	0,728
2 – 3	0,640
3 – 4	0,596
4 – 5	0,583
5 – 6	0,796

NOTE The methods recommended in IEC 60853-1 and IEC 60853-2 uses the load pattern for the last six hours only, on the basis that the thermal effects for load changes prior to that period can, for most electricity supply loads, be represented by the average loss, that is by the loss-load factor.

Loss-load factor for 24 h cycle: 0,504.



Standards Survey

IEC would like to offer you the best quality standards possible. To make sure that we continue to meet your needs, your feedback is essential. Would you please take a minute to answer the questions overleaf and fax them to us at +41 22 919 03 00 or mail them to the address below. Thank you!

Customer Service Centre (CSC)

International Electrotechnical Commission

3, rue de Varembé

1211 Genève 20

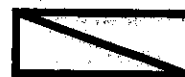
Switzerland

Fax to: IEC/CSC at +41 22 919 03 00

Thank you for your contribution to the standards-making process.

A Prioritaire

Nicht frankieren
Ne pas affranchir



Non affrancare
No stamp required

RÉPONSE PAYÉE

SUISSE

Customer Service Centre (CSC)

International Electrotechnical Commission

3, rue de Varembé

1211 GENEVA 20

Switzerland

Enquête sur les normes

Nous ambitionnons de vous offrir les meilleures normes possibles. Pour nous assurer que nous continuons à répondre à votre attente, nous avons besoin de quelques renseignements de votre part. Nous vous demandons simplement de consacrer un instant à répondre au questionnaire ci-après et de nous le retourner par fax au +41 22 919 03 00 ou par courrier à l'adresse ci-dessous. Merci !

Centre du Service Clientèle (CSC)
Commission Electrotechnique Internationale
3, rue de Varembe
1211 Genève 20
Suisse

Copie: CEI/CSC +41 22 919 03 00

Je vous remercie de la contribution que vous voudrez bien apporter ainsi à la Normalisation Internationale.

A Prioritaire

Nicht frankieren
Ne pas affranchir



Non affrancare
No stamp required

RÉPONSE PAYÉE
SUISSE

Centre du Service Clientèle (CSC)
Commission Electrotechnique Internationale
3, rue de Varembe
1211 GENÈVE 20
Suisse